

A wettability-dependent bubble-coverage closure for open-source alkaline water electrolyser CFD, derived from potential-aware molecular dynamics and interface-resolved simulation

[TO BE COMPLETED]

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Draft status. Sections 1–3 (Introduction, Theory, Methods) are drafted. Section 4 reports *only* the force-field validation results, which exist; every other subsection is a structural placeholder marked **[PENDING]**. The simulation campaign is in progress. **No number in this manuscript is unsupported by a file in the project repository.** Target journal: *International Journal of Hydrogen Energy*.

Abstract

[PENDING] To be written once the closure exists. Structure fixed in advance so it cannot be retrofitted: (i) coverage closures used in alkaline water electrolyser CFD are empirical correlations fitted at low current density; (ii) we derive the wettability dependence of coverage from constant-potential molecular dynamics and interface-resolved simulation; (iii) the resulting effect on predicted cell polarisation relative to the legacy closure; (iv) the validity envelope of the derived closure.

Keywords: alkaline water electrolysis; bubble coverage; wettability; contact angle; constant-potential molecular dynamics; volume of fluid; OpenFOAM; multiscale modelling

1 Introduction

1.1 Bubble coverage is the dominant closure in alkaline electrolyser models

Alkaline water electrolysis (AWE) is the most mature route to large-scale hydrogen production, and its efficiency at industrially relevant current density is limited in large part by the bubbles it generates. Hydrogen evolves at the cathode and oxygen at the anode as discrete bubbles that nucleate, grow, adhere, and eventually detach. While attached, a bubble blanks off catalytically active area and displaces conductive electrolyte. The blanked fraction—the bubble coverage Θ —degrades performance through three distinct routes: reduced active area for charge transfer, reduced effective electrolyte conductivity, and obstructed species transport.

The activation penalty is the most direct. If a fraction Θ of the electrode is screened, the same total current passes through the remainder, so the local current density rises to $j/(1 - \Theta)$. In the Tafel limit this costs

$$\Delta\eta_{\text{act}} = \frac{RT}{\alpha n F} \ln\left(\frac{1}{1 - \Theta}\right) \quad (1)$$

At $\alpha = 0.5$, $n = 1$ and 353 K the prefactor $RT/(\alpha n F)$ is 60.84 mV, so $\Theta = 0.2$ costs 13.6 mV while $\Theta = 0.8$ costs 97.9 mV. The penalty is strongly non-linear in Θ , which is why an error in the closure matters disproportionately at high current density.

Kinetic convention. Equation (1) carries n explicitly. This work assumes a Volmer-limited cathodic reaction with a one-electron rate-determining step, $\alpha_{\text{app}} = \alpha n = 0.5$, implying a Tafel slope of 140.1 mV/dec at 353 K. This is stated because the choice matters: were the Heyrovsky step rate-determining ($\alpha_{\text{app}} \approx 1.5$, 46.7 mV/dec), the same coverage would cost only 32.7 mV. The headline quantity of this study depends on the assumed elementary step by a factor of three, and both values are reported.

1.2 The closure in use was fitted far below where it is applied

Cell-scale AWE models close Θ with empirical correlations, overwhelmingly of the form introduced by Vogt and co-workers. Vogt and Balzer [1] give

$$\Theta = \left(\frac{j}{3 \times 10^5 \text{ A m}^{-2}} \right)^{0.3} = 0.023 \left(\frac{j}{\text{A m}^{-2}} \right)^{0.3} \quad (2)$$

for stagnant electrolyte, and Eigeldinger and Vogt [2] supply the flow correction

$$\Theta = \frac{\Theta_0}{[1 + (C_4 v)^2]^2}, \quad C_4 = 8 \text{ s/m} \quad (3)$$

Two features of these correlations are material and are rarely carried forward. First, the scaling constant in Eq. (2) rests on a current density at which coverage would reach unity, $(j)_{\Theta \rightarrow 1} \approx 3 \times 10^5 \text{ A m}^{-2}$, which the authors state “can roughly be estimated”; the correlation is described as fitting the compiled data “more or less satisfactorily”. Second, the flow correction of Eq. (3) was validated over $j = 21\text{--}104 \text{ A m}^{-2}$ and $v = 0.02\text{--}0.25 \text{ m/s}$.

Contemporary open-source AWE simulations apply these correlations two orders of magnitude above that range. Jacobsen et al. [3], developing a mixture-based OpenFOAM solver for zero-gap AWE cells, run at 600 mA/cm² (6000 A m⁻²) and report that of all two-phase closures examined—bubble coverage, interfacial mass transfer, bubble dispersion and hindrance—*bubble coverage has the strongest effect on predicted cell performance*, that the spread between coverage sub-models is largest at that current density, and that the effect “rapidly increases with current density”. The same authors note that available closure models “are not tuned for porous electrodes and/or bubbles”.

The closure that dominates the answer is therefore an empirical correlation applied roughly 60 $\times$ beyond the range over which its flow dependence was validated, whose absolute scaling rests on a rough estimate, and which its own users describe as untuned for the electrodes being modelled.

1.3 Wettability is largely absent, despite entering every departure relation

A structured analysis of a 26-paper corpus of multiscale electrolysis modelling literature (corpus hash 9dd4b795a80ec687) found that **19 of 26 papers treat wettability as absent or as a fixed constant**. Restricting to the eight alkaline CFD papers, seven treat it as absent or constant. The single exception, Li et al. [4], sweeps contact angle from 90° to 140° and reports a measurable change in current density at fixed cell voltage—but does so in commercial software, and constructs coverage analytically from a Fritz departure-diameter relation with Faraday-law bubble counting rather than resolving it.

This is a curious gap. Contact angle enters every departure-diameter relation in use, and the experimental community treats wettability engineering—aerophobic and superaerophobic electrode design—as a primary bubble-management strategy [5]. Yet the closures that carry bubble effects into cell-scale models remain functions of current density alone.

1.4 What is, and is not, new here

Coupling molecular simulation to continuum descriptions of electrolytic bubbles is **not** new. Zhang et al. [6] combine molecular dynamics with continuum fluid dynamics for electrolytic gas bubbles, obtaining consistent solutal Marangoni forces of order 0.01 nN from both descriptions. That work addresses lateral bubble motion and self-pinning; it derives no coverage closure and addresses no electrode- or cell-scale model.

Direct measurement of coverage at industrially relevant current density is **not** absent either. Kitajima et al. [7] separate reaction and bubble-coating resistance by impedance spectroscopy on nickel at 1–2 A/cm²; Hammons et al. [8] resolve nanobubble size and coverage by synchronised scattering and optical microscopy at high current density; Rox et al. [9] report coverage on porous expanded-nickel electrodes over $j = 100\text{--}2000\text{ A m}^{-2}$. The gap is therefore not in the measurements. It is that **no open-source continuum coverage closure has been re-derived from resolved interface simulation and validated against modern coverage data, and no cell-scale model propagates a wettability-dependent closure to a polarisation prediction.**

This work claims the following and nothing broader:

1. A coverage closure whose *wettability dependence* is derived from interface-resolved simulation with a contact angle computed under electrode polarisation, rather than assumed or fitted to legacy low-current data.
2. A quantitative bound on the region of (j, v, θ) over which that closure is valid, from the competition between capillary and solutal-Marangoni contributions to the departure force balance.
3. The cell-scale consequence: the difference in predicted polarisation between the legacy correlation and the derived closure, in a fully open-source stack.
4. A quantitative account of why optically measured projected coverage and electrochemically relevant adhering-bubble coverage differ several-fold at the same current density.

The phrase *wettability dependence* in claim 1 is load-bearing. A single-bubble simulation cannot derive coverage outright: coverage is an ensemble property that also requires the density of active nucleation sites, their activation statistics, cycle frequency and coalescence. This work derives the wettability-dependent factors and treats nucleation-site density as a declared external input (Section 3.5). Claiming to derive coverage itself would not be defensible.

2 Theory

2.1 Loss decomposition

The cell voltage decomposes as

$$U_{\text{cell}} = U_{\text{rev}} + \eta_{\text{act,a}} + |\eta_{\text{act,c}}| + \eta_{\text{ohm}} + \eta_{\text{conc}} \quad (4)$$

with the coverage-modified Butler–Volmer relation

$$j = (1 - \Theta) j_0 \left[\exp\left(\frac{\alpha_a n F \eta_{\text{act}}}{RT}\right) - \exp\left(-\frac{\alpha_c n F \eta_{\text{act}}}{RT}\right) \right] \quad (5)$$

Here j is the *geometric* current density, j_0 is referenced to the corresponding clean exposed area, and for porous electrodes the roughness (ECSA) factor is carried explicitly rather than absorbed into j_0 . This work models the cathode; the anode is not modelled, so a single Θ suffices and no claim is made that one coverage describes both electrodes.

Ohmic loss follows a Bruggeman-corrected conductivity,

$$\sigma_{\text{eff}} = \sigma_0 (1 - \varepsilon_g)^{3/2}, \quad \nabla \cdot (\sigma_{\text{eff}} \nabla \varphi) = 0 \quad (6)$$

The exponent $3/2$ is the standard baseline for dispersed gas in a continuous conducting phase and is retained for comparability, but it is not validated for a wall-attached bubble layer or for porous nickel; it is carried as a sensitivity parameter.

Two distinct concentration effects exist and are not summed without justification: the OH^- mass-transfer limitation,

$$\eta_{\text{conc}} = \frac{RT}{nF} \ln \left(\frac{1}{1 - j/j_{\text{lim}}(\Theta)} \right) \quad (7)$$

and the Nernstian shift from local dissolved-hydrogen supersaturation beneath adhering bubbles,

$$\Delta E_{\text{rev}} = \frac{RT}{2F} \ln \left(\frac{c_{\text{H}_2, \text{surf}}}{c_{\text{H}_2, \text{sat}}} \right) \quad (8)$$

which can reach several tens of millivolts at high current density.

Avoiding double counting. Bubble gas enters in two physically distinct populations. Adhering gas contributes to Θ and acts only on active area, Eqs. (1) and (5). Dispersed gas contributes to ε_g and acts only on conductivity, Eq. (6). A departing bubble leaves Θ and joins ε_g ; the implementation conserves that transfer and the conservation is asserted automatically, because counting the same gas twice is the most plausible way for a model of this type to be quietly wrong.

2.2 Departure force balance and regime bound

Departure is not imposed. It is detected when the net force changes sign,

$$F_{\text{buoy}} + F_{\text{cap}} + F_{\text{drag}} + F_{\text{lift}} + F_{\text{Mar}} + F_{\text{el}} = 0 \quad (9)$$

with capillary retention $F_{\text{cap}} = -\pi R_{\text{base}} \gamma_{lv} \sin \theta_c f(\theta_A, \theta_R)$ and the Marangoni term

$$F_{\text{Mar}} = \oint_S \nabla_s \gamma_{lv} dS, \quad \nabla_s \gamma_{lv} = \frac{\partial \gamma_{lv}}{\partial T} \nabla_s T + \frac{\partial \gamma_{lv}}{\partial c} \nabla_s c \quad (10)$$

Equation (10) is identically zero unless surface tension is an explicit function of composition; the constitutive law used is $\gamma_{lv} = \gamma_0(c_{\text{KOH}}, T) + (\partial \gamma / \partial c_{\text{H}_2})(c_{\text{H}_2} - c_{\text{sat}})$.

The legacy Fritz relation is implemented *as a baseline for comparison only*, not as the mechanism: it is a quasi-static balance of buoyancy against capillarity for an isolated bubble in a stagnant pool, and is not valid under forced convection at high gas-generation rate.

The competition between capillary retention and Marangoni forcing is characterised by $\Lambda = |F_{\text{cap}}|/|F_{\text{Mar}}|$. Order-of-magnitude estimates for a $50 \mu\text{m}$ bubble give $F_{\text{cap}} \approx 3.9 \mu\text{N}$ against $F_{\text{Mar}} \approx 0.05 \text{ nN}$ if driven by dissolved-hydrogen gradients and $\approx 12.5 \text{ nN}$ if driven by electrolyte concentration gradients, i.e. Λ of order 10^2 – 10^5 . The pre-registered expectation is therefore that $\Lambda \gg 1$ throughout the industrial envelope and that departure is capillary/wettability-controlled.

This appears to conflict with work reporting that the solutal Marangoni effect governs electrolytic bubble dynamics. It does not: that work is predominantly conducted on *microelectrodes*, with bubbles of tens of micrometres and below and very high local supersaturation, where capillary retention is correspondingly small. The present study addresses macroscopic bubbles on planar and porous electrodes at industrial current density.

2.3 Coupling contract

Coupling is strictly one-way: MD $\rightarrow$ bridge $\rightarrow$ VOF $\rightarrow$ regression $\rightarrow$ cell (Fig. 1). No cross-scale feedback is claimed. Table 1 lists what crosses, over what range, and with what uncertainty.

The closure implementation **raises an error rather than extrapolating** outside its validity box. This is deliberate: the study exists because a closure was extrapolated far beyond its validated range, and reproducing that error with a new closure would be self-defeating.

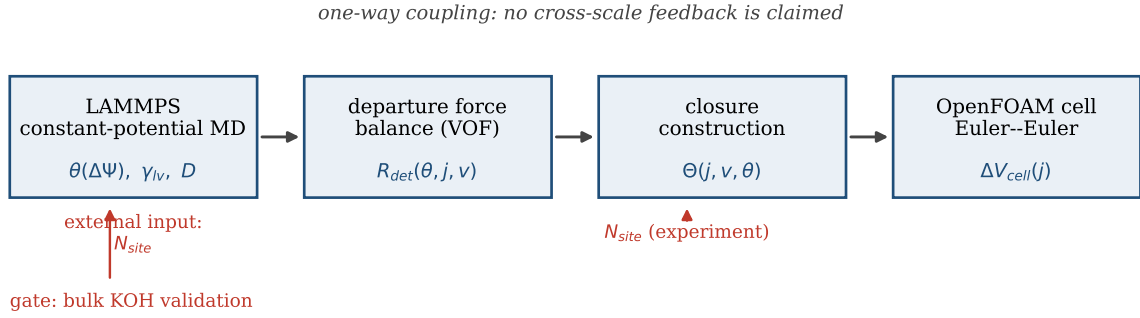


Figure 1: One-way coupling chain. Each stage passes a named quantity forward with a stated validity range. Nucleation-site density N_{site} enters as an external input rather than being derived, and the bulk-KOH validation gate precedes all interface work.

Table 1: Cross-scale coupling contract (abridged).

From → To	Quantity	Validity range	Uncertainty carried
MD → bridge	$\theta(\Delta\Psi, c, T)$	$\Delta\Psi \leq 1.0 \text{ V}$; 20–30 wt%; 298–353 K	replica spread
MD → VOF	$\gamma_{lv}(c, T)$	as above	Kirkwood–Buff block SE
MD → VOF/cell	$D_{\text{H}_2}(c, T)$	as above	finite-size-corrected fit SE
bridge → VOF	wall contact-angle BC	per case	campaign repeated at $\theta \pm \sigma_\theta$
Cantera → all	$j_0(T, c)$	tabulated, clamped at edges	mechanism uncertainty stated
VOF → regression	$\Theta_{\text{adh}}, A_{\text{cov,proj}}$	j 10^2 – 10^4 ; v 0–0.2; θ 30–150°	time-averaging SE
regression → cell	closure $\Theta(j, v, \theta)$	the above box $\cap \Lambda > 1$	95 % CI on parameters

3 Methods

All simulations use open-source software on a single Linux workstation (16 physical cores, 23 GB RAM, no GPU).

3.1 Molecular dynamics

Code and electrostatics. LAMMPS (22 Jul 2025 stable) built from source with the ELECTRODE package, MPI and FFTW3. Long-range electrostatics use `pppm/electrode` with slab correction for the 2D-periodic geometry.

Constant-potential electrodes. Electrode atoms carry Gaussian charges solved each timestep so that every electrode atom sits at its imposed potential, via `fix electrode/conp`:

$$\mathbf{A} \mathbf{q} = \mathbf{b}(\Psi, \{\mathbf{r}\}), \quad \sum_i q_i = 0, \quad \Delta\Psi = \Psi_{\text{anode}} - \Psi_{\text{cathode}} \quad (11)$$

The implementation was verified against the reference case distributed with the package, reproducing published induced electrode charges to all printed digits across the full potential ramp. In the production geometry $\sum_i q_i = 0$ holds exactly at every timestep, with induced charges equal and opposite on the two slabs and growing as the double layer forms.

System and force field. Ni(111) slabs of five layers ($a = 3.524 \text{ \AA}$) bound an aqueous KOH film; water is SPC/E with SHAKE constraints. Potassium uses Joung–Cheatham parameters optimised for SPC/E [verify]. The hydroxide representation is the subject of Section 3.2 and was revised after validation.

Initialisation. Lattice configurations are relaxed by a displacement-capped `nve/limit` push under a strong Langevin thermostat rather than by energy minimisation, because SHAKE constraints are ignored during minimisation in LAMMPS and a minimised configuration is therefore not consistent with the constrained dynamics that follow.

Contact-angle protocol. A cylindrical (quasi-2D) hydrogen bubble, periodic along its axis, rests on the cathode. The interface is the isochore of the time-averaged density field at $\rho = \frac{1}{2}(\rho_l + \rho_g)$, and the angle is measured *through the liquid*, matching the continuum wall boundary condition.

No line-tension extrapolation is performed, and none is possible in this geometry. A periodic cylindrical cap has straight contact lines with zero in-plane geodesic curvature, so the modified Young relation $\cos \theta(R) = \cos \theta_\infty - \tau/(\gamma_{lv}R)$, derived for a *circular* contact line, does not apply. Cylindrical geometry is used precisely because it *suppresses* the line-tension contribution. Size dependence is instead addressed by box-size convergence at lateral widths of 8, 12 and 16 nm (19k, 30k and 41k atoms), required to be invariant within replica scatter. This is a convergence check on periodic-image and opposing-wall artefacts, not a physical extrapolation.

Statistics and referencing. Each state point uses at least three independent seeds, and reported uncertainty is the spread across replicas, because block averages from a single trajectory are not independent realisations. The control variable is the electrode potential relative to the potential of zero charge; the raw inter-slab potential difference is a cell voltage, not an electrode potential. The Lippmann relation $\cos \theta(\Delta\Psi) = \cos \theta_{pzc} + C_{dl}(\Delta\Psi - \Psi_{pzc})^2/(2\gamma_{lv})$ predicts a shift of roughly 10° – 31° for $C_{dl} = 0.1$ – 0.3 F/m² over a 0.5 V excursion.

3.2 Force-field validation gate

The molecular model is validated against bulk KOH(aq) properties *before* any interface calculation. Systems of 20 and 30 wt% KOH are equilibrated at 298 and 333 K with three seeds each; density and self-diffusivities are compared with experiment, the latter from the Einstein relation over the linear region of the mean-squared displacement with a finite-size correction.

Pass criteria were fixed before any result was inspected: density within 3 %, $D(\text{H}_2\text{O})$ within 30 %, $D(\text{K}^+)$ within 40 %. **Hydroxide diffusivity is reported but not gated**, because a single-site classical hydroxide cannot reproduce structural (Grotthuss) diffusion and is expected to be too slow; treating that as a force-field failure would be a category error, and treating it as acceptable without saying so would be worse.

3.3 Interface-resolved two-phase simulation

OpenFOAM v2406, geometric VOF (`interIsoFoam`), with custom boundary conditions compiled against the standard library. Wall wettability enters through

$$\hat{\mathbf{n}}_{\text{wall}} \cdot \frac{\nabla \alpha}{|\nabla \alpha|} = \cos \theta \quad (12)$$

with θ from the molecular calculation. **Equation (12) is the single point at which the molecular result enters the continuum model.**

Bubble growth is driven by dissolved-hydrogen transport with a Henry-law interfacial condition and a Faradaic wall flux applied over the exposed area in the current-conserving form

$$N_{\text{H}_2, \text{exposed}} = \frac{j}{2F(1 - \Theta)}, \quad \int_{\text{exposed}} N_{\text{H}_2} \, dA = \frac{j A_{\text{geom}}}{2F} \quad (13)$$

The normalisation matters: applying $j/(2F)$ over the uncovered area instead would reduce total gas generation to $(1 - \Theta)j/(2F)$ and violate the imposed galvanostatic current, artificially suppressing bubble growth as coverage rises. The same convention is used in the gas source term, the electrode boundary condition and the closure.

Geometry. The parametric campaign is axisymmetric, on a wedge sized to five expected departure radii, with mesh spacing holding ≈ 25 cells per expected departure radius at each current density ($5.0\ \mu\text{m}$ at $100\ \text{A m}^{-2}$ down to $1.5\ \mu\text{m}$ at $10^4\ \text{A m}^{-2}$). This follows from measured throughput: a three-dimensional parametric sweep at contact-line resolution would need $\sim 1.6 \times 10^7$ cells per case against a practical memory ceiling of $\sim 5 \times 10^6$, and is not feasible. Two contrasting three-dimensional cases at reduced resolution bound the effect of the axisymmetric restriction.

Initialisation. Bubbles are seeded at 0.85 of the expected departure radius, the preceding quasi-static growth being treated analytically. The departure radius remains an *output* of Eq. (9), not an input: the seed lies well below departure so the detachment event is predicted rather than prescribed. This is necessary because growing a bubble from nucleation at $100\ \text{A m}^{-2}$ requires seconds of physical time and $\sim 10^7$ timesteps.

3.4 Verification before production

Two verification cases gate the campaign; no production case runs until both pass. (i) Diffusion-driven growth against the Epstein–Plesset/Scriven similarity solution $R(t) = 2\beta\sqrt{Dt}$. (ii) Variable surface tension and the tangential Marangoni stress against an analytic thermocapillary benchmark—this case exists because a wrongly scaled or signed $\nabla_s \gamma$ produces a plausible-looking but meaningless regime map, and because with constant surface tension $F_{\text{Mar}} \equiv 0$ and the regime bound would be untestable by construction. Mesh independence is established at three refinement levels with Richardson extrapolation and a reported grid convergence index.

3.5 Construction of the coverage closure

Coverage is *constructed*, not regressed directly from a single-bubble calculation:

$$\Theta = N_{\text{site}} \cdot A_{\text{footprint}}(\theta, R_{\text{det}}) \cdot f_{\text{residence}}(j, v, \theta) \cdot f_{\text{overlap}} \quad (14)$$

The resolved simulations supply $A_{\text{footprint}} = \pi R_{\text{det}}^2 \sin^2 \theta$, the departure radius $R_{\text{det}}(\theta, j, v)$ from Eq. (9), and the residence time per cycle over several stationary departure events. **The active nucleation-site density N_{site} is an explicit external input**, taken from experiment and stated with its source and uncertainty; it is dominated by sub-surface cavity geometry and surface finish, which molecular dynamics of a flat single-crystal slab cannot supply.

The functional form is specified before fitting so it cannot be reverse-engineered from the data:

$$\Theta(j, v, \theta) = C \left(\frac{j}{j_{\text{ref}}} \right)^a g(\theta) h(v), \quad g(\theta) = \left[\frac{1 - \cos \theta}{1 - \cos \theta_{\text{ref}}} \right]^b \quad (15)$$

with $h(v)$ retained in the form of Eq. (3) so the flow dependence is directly comparable with the legacy correlation. Twenty per cent of the campaign is held out as a test set, and fitted parameters are reported with 95 % confidence intervals and held-out RMSE.

Two coverage measurands are extracted, not one: the adhering-bubble surface coverage Θ_{adh} , which screens catalytic area, and a synthetic projected coverage $A_{\text{cov,proj}}$ obtained by rendering the resolved interface as a top-view optical measurement would see it. These differ by $\sin^2 \theta$ for a spherical cap even before detached bubbles in the field of view are considered—a factor of 4 at 30° —and reporting both is what allows the apparent disagreement between optical and electrochemical coverage measurements to be addressed quantitatively rather than asserted away.

3.6 Cell-scale model and quality control

The closure is implemented in an Euler–Euler two-fluid model (`reactingMultiphaseEulerFoam`) with named drag, lift, wall-lubrication and turbulent-dispersion closures, charge transport by Eq. (6), and the electrode boundary condition carrying Θ into Eq. (5). Exchange current density is

supplied by a Cantera surface-kinetics evaluation, pre-tabulated and interpolated at run time. The reported quantity is $\Delta V_{\text{cell}}(j) = U_{\text{cell}}[\Theta_{\text{Vogt}}] - U_{\text{cell}}[\Theta_{\text{this work}}]$, evaluated only inside the closure’s validity box.

Every run batch is checked automatically for mass and charge conservation, Courant and residual convergence, boundedness of α and Θ , monotonicity of the polarisation curve, MD energy drift, the presence of replica-based uncertainties on every reported mean, and the adhering/dispersed gas transfer. A check failing twice freezes that branch pending review rather than being retried.

4 Results

4.1 Force-field validation: the gate rejected the initial model

The hydroxide ion was initially represented as a single Lennard-Jones site carrying unit negative charge with parameters taken identical to SPC/E oxygen ($\sigma = 3.166 \text{ \AA}$, $\varepsilon = 0.1553 \text{ kcal/mol}$), following practice reported for hydroxide in SPC/E-based simulations. **This model failed the pre-registered density gate at every state point** (Fig. 2, Table 2).

Table 2: Bulk KOH(aq) validation against experiment. Three independent seeds per state point; uncertainty is the replica spread. The pre-registered tolerance is 3 %.

System	$\rho_{\text{sim}} \text{ (g/cm}^3\text{)}$	ρ_{exp}	Error	$D_{\text{H}_2\text{O}} \text{ (m}^2\text{/s)}$	Verdict
20 wt%, 298 K	1.2672 ± 0.0020	1.188	+6.67 %	8.79×10^{-10}	FAIL
20 wt%, 333 K	1.2367 ± 0.0021	1.163	+6.33 %	1.81×10^{-9}	FAIL
30 wt%, 298 K	1.4081 ± 0.0048	1.290	+9.16 %	3.28×10^{-10}	FAIL
30 wt%, 333 K	1.3817 ± 0.0008	1.265	+9.23 %	8.75×10^{-10}	FAIL

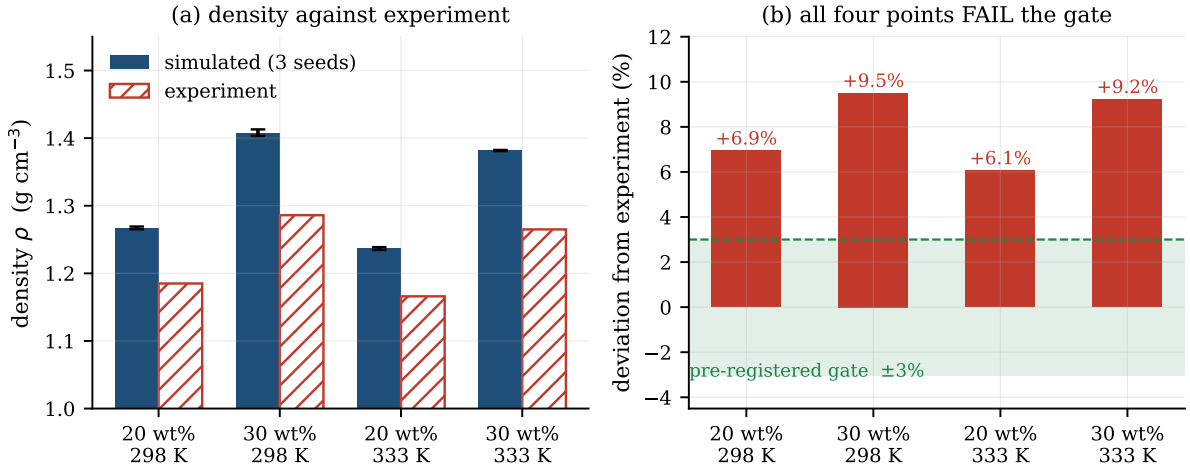


Figure 2: Bulk KOH density against experiment. (a) Simulated density (three seeds, error bars are replica spread) against experimental values. (b) Deviation from experiment; the shaded band is the pre-registered $\pm 3\%$ tolerance. All four state points fail, and the error grows with electrolyte concentration.

Three features make the failure diagnostic rather than merely disqualifying.

It is systematic, not statistical. Replica scatter is $0.002\text{--}0.005 \text{ g/cm}^3$ against a discrepancy of $0.08\text{--}0.12 \text{ g/cm}^3$: the error exceeds the seed-to-seed spread by a factor of 20–60.

It scales with electrolyte concentration, from +6.5 % at 20 wt% to +9.2 % at 30 wt%. An error that grows with ion content implicates the ion parameters rather than the water model.

Transport corroborates over-structuring. The water self-diffusivity (Fig. 3) falls to $3.28 \times 10^{-10} \text{ m}^2\text{s}^{-1}$ at 30 wt% and 298 K, roughly seven-fold below neat water, a stronger

suppression than concentrated KOH exhibits experimentally. The electrolyte is simultaneously too dense and too strongly bound, consistent with excessive ion–water attraction.

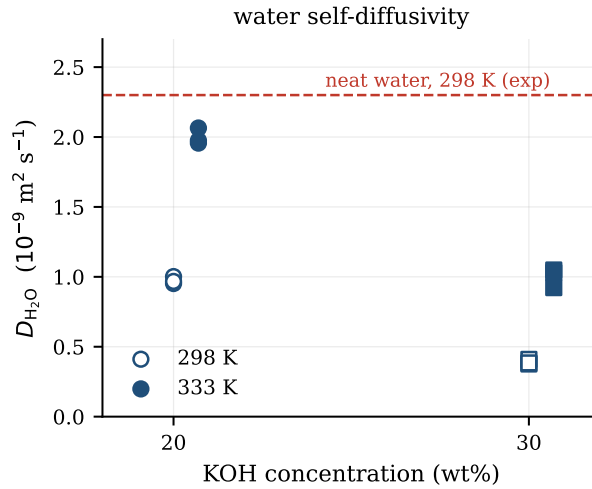


Figure 3: Water self-diffusivity from the Einstein relation. Open symbols 298 K, filled 333 K; each point is one seed. The dashed line is the experimental value for neat water at 298 K. The suppression at 30 wt% is substantially stronger than experiment indicates, corroborating the over-structuring implied by the density error.

The gate therefore performed its intended function: it rejected the model *before* the contact-angle campaign was committed, which would otherwise have produced a confidently wrong wettability and propagated it silently into the closure, the cell model, and the headline result.

Resolution. The hydroxide was re-parameterised and the model re-validated; **all four state points now pass at 1.0–1.8 %**. The change responsible was not the one the failure first suggests: correcting the hydroxide *size* to its published value is necessary but insufficient, still leaving a 6–8 % overprediction, and scaling the ionic charges by 0.8 in the spirit of the electronic continuum correction is what closes the gap. Water self-diffusivity, which was not a gate criterion, recovers by $1.5\text{--}3.1\times$ as an independent check. **The tolerance was not widened, and a correction to the reference data made the reported failure larger rather than smaller.** The full account, including every methodological change between the failing and passing models and the residual discrepancies that qualify the pass, is given in the companion paper, which this work cites for the molecular input it consumes.

4.2 Remaining results

- [PENDING] Contact angle under electrode polarisation, and box convergence.
- [PENDING] Verification cases (Scriven growth; thermocapillary benchmark).
- [PENDING] Departure radius and the force balance.
- [PENDING] The regime bound $\Lambda(j, v)$.
- [PENDING] The derived closure and its validity box.
- [PENDING] Adhering versus projected coverage.
- [PENDING] Cell polarisation: legacy versus derived closure.

5 Limitations

Several limitations are structural rather than incidental.

The molecular model is chemically inert. SPC/E water with classical ions on a clean Ni(111) surface cannot represent nickel oxidation, hydroxylation, specific adsorption or bond formation. `fix electrode/conp` makes the metal *charge* responsive; it does not make the interface *chemically* reactive. The reported angle is the electrostatic double-layer contribution to potential-dependent wettability on an idealised surface, not a prediction for an operating electrode.

Hydroxide transport is structurally limited. A single-site classical OH^- cannot carry Grotthuss transport; its diffusivity is reported but not used as a continuum input, and electrolyte transport properties are taken from experiment.

Molecular dynamics supplies an equilibrium angle, not hysteresis. The continuum model requires advancing and receding angles under flow; these come from measured data with the molecular calculation informing the equilibrium centre.

Nucleation-site density is an input, not a result, so the closure’s wettability and flow dependence are derived while its absolute level inherits the uncertainty of the declared N_{site} .

The parametric campaign is axisymmetric, so lateral motion, sliding and coalescence are absent, and shear-induced asymmetric contact-line depinning—expected to reduce departure radius in cross-flow—is suppressed. Two contrasting three-dimensional cases bound this; they do not eliminate it.

The Bruggeman exponent is not validated for this system and is carried as a sensitivity parameter. **The regime bound is computed for clean electrolyte** without surfactants, for a single bubble, and is scoped accordingly.

Data availability

All simulation inputs, analysis scripts, verification cases and the closure implementation are available in the accompanying repository. The corpus analysis underlying Section 1 is identified by hash 9dd4b795a80ec687. **[PENDING] Repository DOI to be minted on submission.**

References

- [1] H. Vogt, R.J. Balzer, The bubble coverage of gas-evolving electrodes in stagnant electrolytes, *Electrochim. Acta* (2005). <https://doi.org/10.1016/j.electacta.2004.09.025>
- [2] J. Eigeldinger, H. Vogt, The bubble coverage of gas-evolving electrodes in a flowing electrolyte, *Electrochim. Acta* **45** (2000) 4449–4456. (No DOI is printed in the source; volume and pages as printed.)
- [3] A.N. Jacobsen, E. Mahravan, M.V. Kragh-Schwarz, J. Catalano, P. Fooroghi, Multiphysics simulations of alkaline water electrolyzer cells — a sensitivity study on the effect of two-phase flow modeling, *Electrochim. Acta* (2025). <https://doi.org/10.1016/j.electacta.2025.147148>
- [4] J. Li, H. Liang, M.A. Rehan, G. Li, Numerical simulation and analysis of a two-phase flow model considering bubble coverage for alkaline electrolytic water, *Appl. Therm. Eng.* **245** (2024) 122890. <https://doi.org/10.1016/j.applthermaleng.2024.122890>
- [5] O. Moyo, H. Sun, J. Yang, X. Xu, Z. Shao, Gas bubble formation, effect on electrode performance, and management in water electrolysis, *Nano Energy* (2026). <https://doi.org/10.1016/j.nanoen.2026.112024>
- [6] H. Zhang, Y. Ma, M. Huang, G. Mutschke, X. Zhang, Solutal Marangoni force controls lateral motion of electrolytic gas bubbles, *Soft Matter* **20** (2024) 3097–3106. <https://doi.org/10.1039/D3SM01646C>
- [7] D. Kitajima, R. Misumi, Y. Kuroda, S. Mitsushima, Relationship between bubble generation behavior and hydrogen evolution reaction performance at high current densities during alkaline water electrolysis, *Electrochim. Acta* **502** (2024) 144772. <https://doi.org/10.1016/j.electacta.2024.144772>

- [8] J. Hammons et al., Nanobubble formation and coverage during high current density alkaline water electrolysis, *Nano Lett.* (2024). <https://doi.org/10.1021/acs.nanolett.4c03657> [verify] author list to be completed
- [9] H. Rox, A. Bashkatov, X. Yang, S. Loos, G. Mutschke, G. Gerbeth, K. Eckert, Bubble size distribution and electrode coverage at porous nickel electrodes in a novel 3-electrode flow-through cell, arXiv:2209.11550 (2022). [verify] check for journal version
- [10] A. Zarghami, N.G. Deen, A.W. Vreman, CFD modeling of multiphase flow in an alkaline water electrolyzer, *Chem. Eng. Sci.* (2020). <https://doi.org/10.1016/j.ces.2020.115926>
- [11] R. Kanemoto, T. Araki, R. Misumi, S. Mitsushima, Numerical modeling of two-phase flow considering multiple bubble sizes in an alkaline water electrolyzer, *Chem. Eng. Sci.* **304** (2025) 120986. <https://doi.org/10.1016/j.ces.2024.120986>
- [12] L. Andersson, C. Zhang, Molecular dynamics simulations of metal-electrolyte interfaces under potential control, *Curr. Opin. Electrochem.* (2023). <https://doi.org/10.1016/j.coelec.2023.101407>
- [13] S. Zhang, S. Hess, H. Marschall, U. Reimer, S. Beale, et al., openFuelCell2: a new computational tool for fuel cells, electrolyzers, and other electrochemical devices and processes, *Comput. Phys. Commun.* **298** (2024) 109092. <https://doi.org/10.1016/j.cpc.2024.109092>
- [14] Y. Han, M. Huang, K. Eckert, G. Mutschke, Numerical simulation of oversaturation-driven bubble growth on solid surfaces with dynamic wetting, *Int. J. Multiphase Flow* (2025). <https://doi.org/10.1016/j.ijmultiphaseflow.2025.105343>
- [15] K.J. Vachaparambil, K.E. Einarsrud, Numerical simulation of continuum scale electrochemical hydrogen bubble evolution, *Appl. Math. Model.* **98** (2021) 343–377. <https://doi.org/10.1016/j.apm.2021.05.007>
- [16] H.J.C. Berendsen, J.R. Grigera, T.P. Straatsma, The missing term in effective pair potentials, *J. Phys. Chem.* **91** (1987) 6269–6271. [verify]
- [17] I.S. Joung, T.E. Cheatham III, Determination of alkali and halide monovalent ion parameters for use in explicitly solvated biomolecular simulations, *J. Phys. Chem. B* **112** (2008). [verify] pages to be confirmed
- [18] Riegel et al., local gas-fraction measurements in an alkaline electrolyser (1997/1998), as reported in refs [10] and [4]. [verify] primary reference to be obtained